University of Sharjah
College of Computing and Informatics
Computer Engineering Department
Senior Design Project
(Please Modify ALL RED colored text)

Project Number	Project Title : **Design and Implementation of an Edge Based Driver Monitoring System with False Alarm Reduction**
SDP1-34	**العنوان باللغة العربية تصميم وتنفيذ نظام مراقبة للسائق قائم على الحوسبة الطرفية مع الحد من الإنذارات الخاطئة**

Term: Fall **Year: 2026-2027**

Students Details:

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Project Specialty (PLEASE TICK UP TO TWO CHOICES ONLY):

CyberSecurity/Software	
Hardware/embedded systems	X
AI/Robotics	X
General Computer	

Supervisor(s) Details

Full Name: Mohamad Khairi Bin Ishak Signature: Signed by email? (**YES**/NO)

1- **Problem statement** (Maximum 150 words)

As countries increasingly mandate the usage of in-cabin Driver Monitoring Systems (DMS) [1] aiming to minimize accidents caused by driver drowsiness and distraction, edge deployment of lightweight vision models has become essential. However, edge system deployment faces three core bottlenecks:

1. False detections can trigger unnecessary alerts and downstream operations, potentially reducing driver trust and system reliability.
2. Real time monitoring requires low inference latency, while limited computational resources, memory, power, and thermal capacity restrict the use of computationally intensive models on edge devices.
3. Variations in cabin conditions and visible (RGB) and Near Infrared (NIR) illumination can affect model performance and increase susceptibility to false detections.

This introduces a clear tradeoff: Higher compute systems effectively minimize false detections, yet their footprint renders them infeasible in real-time deployment.

Overall, this project addresses this challenge through lightweight object detection, hard negative mining, and embedded system implementation.

2- **Briefly describe the project summary, importance and Objectives** (Maximum 150 words)

Drowsiness and distraction are major factors of road crashes and contribute to more than 35% of road fatalities [2]. While in-cabin Driver Monitoring Systems (DMS) help to reduce these risks, these systems frequently produce nuisance false alarms, which undermine driver trust, encouraging the driver to circumvent the monitoring system[3].

The project has three main objectives:
(1) to evaluate lightweight object detection models using RGB and NIR driver monitoring datasets
(2) to investigate negative sample ratios and hard negative mining for reducing false detections while maintaining detection performance
(3) to deploy the selected model on a Raspberry Pi based prototype integrating a camera and alert mechanism. The final system will be evaluated using detection accuracy, F1 score, false detections per 1,000 negative frames, inference latency, and frame rate. The project combines AI model development, dataset optimization, embedded implementation, and real time system evaluation.

3- **Briefly describe the project methodology** (Maximum 150 words)

We follow a three-phase engineering methodology to realize the edge DMS. First, we benchmark nano-scale (<5M parameters) detection families, such as DFINE and YOLO, across RGB and NIR in-cabin 30FPS video datasets such as (DMD and Drive&Act). Models are evaluated on localized (not temporal) inattention cues such as (yawning, eating, drinking) using an equally-weighted normalized score (mAP:50, macro-F1, FPS, false detections per 1,000 frames) to render the optimal model that balances accuracy, detection and latency. Second, we calibrate the dataset across negative-to-positive frame ratios such as (0%,20%,40%,60%) and curate the dataset with hard false positives instead of random negative frames to suppress false detections.

Finally, the selected lightweight model will be optimized and deployed on a Raspberry Pi based edge platform integrated with an RGB camera and real time alert mechanism. The complete embedded system will be experimentally evaluated in terms of inference speed, processing latency, detection reliability, false alarm rate, and overall real time performance under representative driver monitoring conditions.

4- **Briefly describe the expected outcomes** (Tick one or more of the following choices)

What are the outcomes of your project: e.g. software, prototype that does so and so, simulation to prove research points, conference paper, etc. Write a paragraph and then select from the table below.

Outcome	Choice
Physical Prototype	**X**
Simulation demo	
Research paper	**X**
Software app/module	

The project delivers three main outcomes. Firstly, a report covering the literature review, dataset curation methodology and embedded deployment results. Second, an optimized lightweight object detection model for edge-based Driver Monitoring Systems, trained using curated data to reduce false detections while maintaining reliable detection performance. Third, a functional embedded prototype implemented on a Raspberry Pi 4, integrating a real time RGB camera, edge AI inference, and an alert mechanism for in cabin driver monitoring. The prototype will demonstrate the practical feasibility of deploying lightweight AI models for reliable and real time driver monitoring on resource constrained edge hardware.

References:

[1] F. Lyrheden, "How Driver Monitoring Systems (DMS) Are Being Made Mandatory in 18 Million European Cars," Smart Eye, Apr. 28, 2023. [Online]. Available: https://smarteye.se/blog/the-general-safety-regulations-gsr-and-driver-monitoring-systems-dms/. [Accessed: Sep. 6, 2026].
[2] G. Merlhiot and M. Bueno, "How drowsiness and distraction can interfere with take-over performance: A systematic and meta-analysis review," *Accident Analysis & Prevention*, vol. 170, Art. no. 106536, Jun. 2022, doi: 10.1016/j.aap.2021.106536.

[3] F. Lambert, "Chinese drivers are using plastic heads to fool Tesla's Autopilot cabin camera," *Electrek*, Jun. 15, 2026. [Online]. Available: https://electrek.co/2026/06/15/chinese-drivers-plastic-heads-fool-tesla-autopilot-camera/. [Accessed: Sep. 6, 2026].